

BPL_TEST2_Batch - demo

```
In [1]: # Import the application model and context variables
        from BPL_TEST2_Batch_fmpy import*
```

Windows - run FMU pre-compiled JModelica 2.14

```
In [2]: # Import the general FMU_explore functions ana adapt them to the application
        from fmu_explore_fmpy import AdaptWith

        Adapted = AdaptWith(parValue, parLocation, parCheck, keyVariables, \
                               model_description, fmu_model, fmu_process_diagram, \
                               MSL_usage, MSL_version, BPL_version, \
                               opts_std, simulationTime, timeDiscreteStates, \
                               diagrams, ax, lines)

        par = Adapted.par
        init = Adapted.init
        disp = Adapted.disp
        simu = Adapted.simu
        show = Adapted.show

        resetPen = Adapted.resetPen
        process_diagram = Adapted.process_diagram
        describe_general = Adapted.describe_general
        describe_parts = Adapted.describe_parts
        describe_MSL = Adapted.describe_MSL
        FMU_explore_info = Adapted.FMU_explore_info
        system_info = Adapted.system_info
        SDG = Adapted.SDG
```

```
In [3]: run -i BPL_TEST2_Batch_fmpy_explore.py
```

Model for the process has been setup. Key commands:

- par() - change of parameters and initial values
- init() - change initial values only
- simu() - simulate and plot
- newplot() - make a new plot
- show() - show plot from previous simulation
- disp() - display parameters and initial values from the last simulation
- describe() - describe culture, broth, parameters, variables with values/units

Note that both disp() and describe() takes values from the last simulation and the command process_diagram() brings up the main configuration

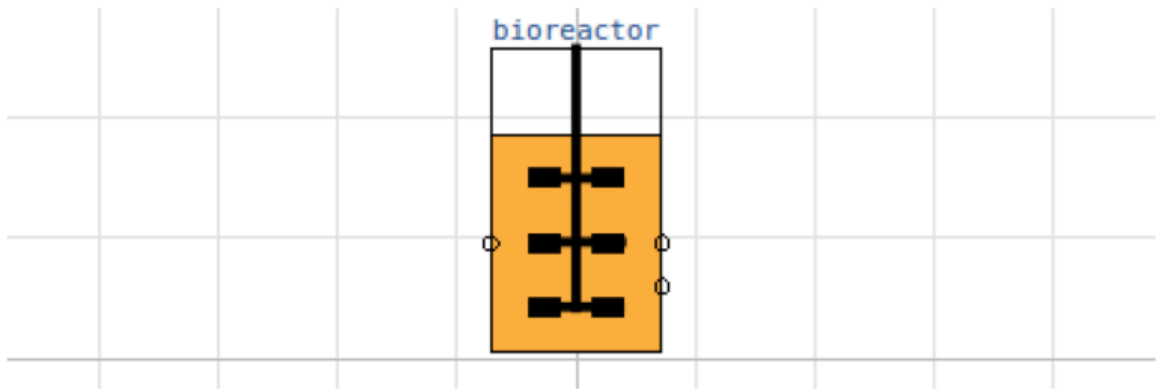
Brief information about a command by help(), eg help(simu)

Key system information is listed with the command system_info()

```
In [4]: %matplotlib inline
        plt.rcParams['figure.figsize'] = [25/2.54, 20/2.54]
```

```
In [5]: process_diagram()
```

No processDiagram.png file in the FMU, but try the file on disk.



```
In [6]: describe('process')
```

None

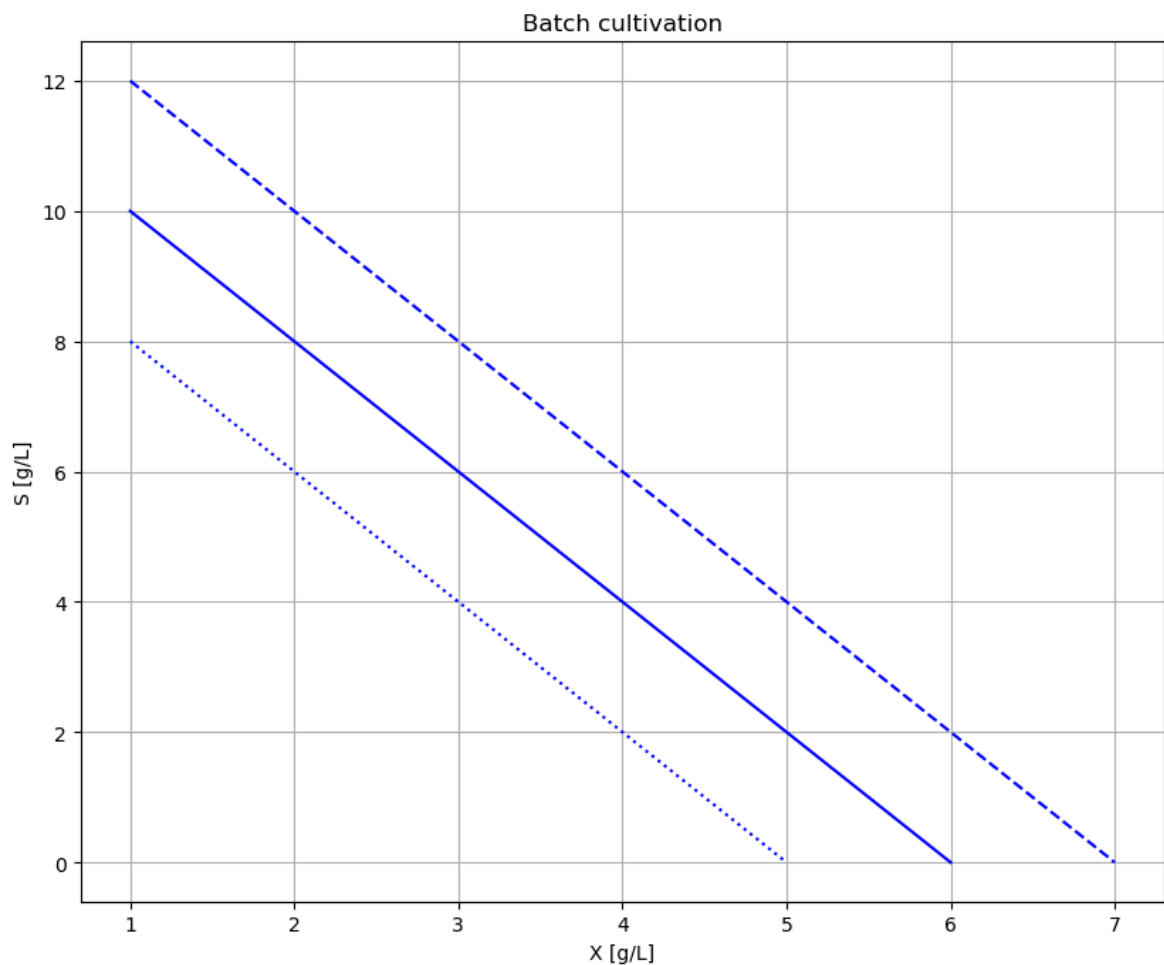
```
In [7]: print(read_model_description(fmu_model).description)
```

None

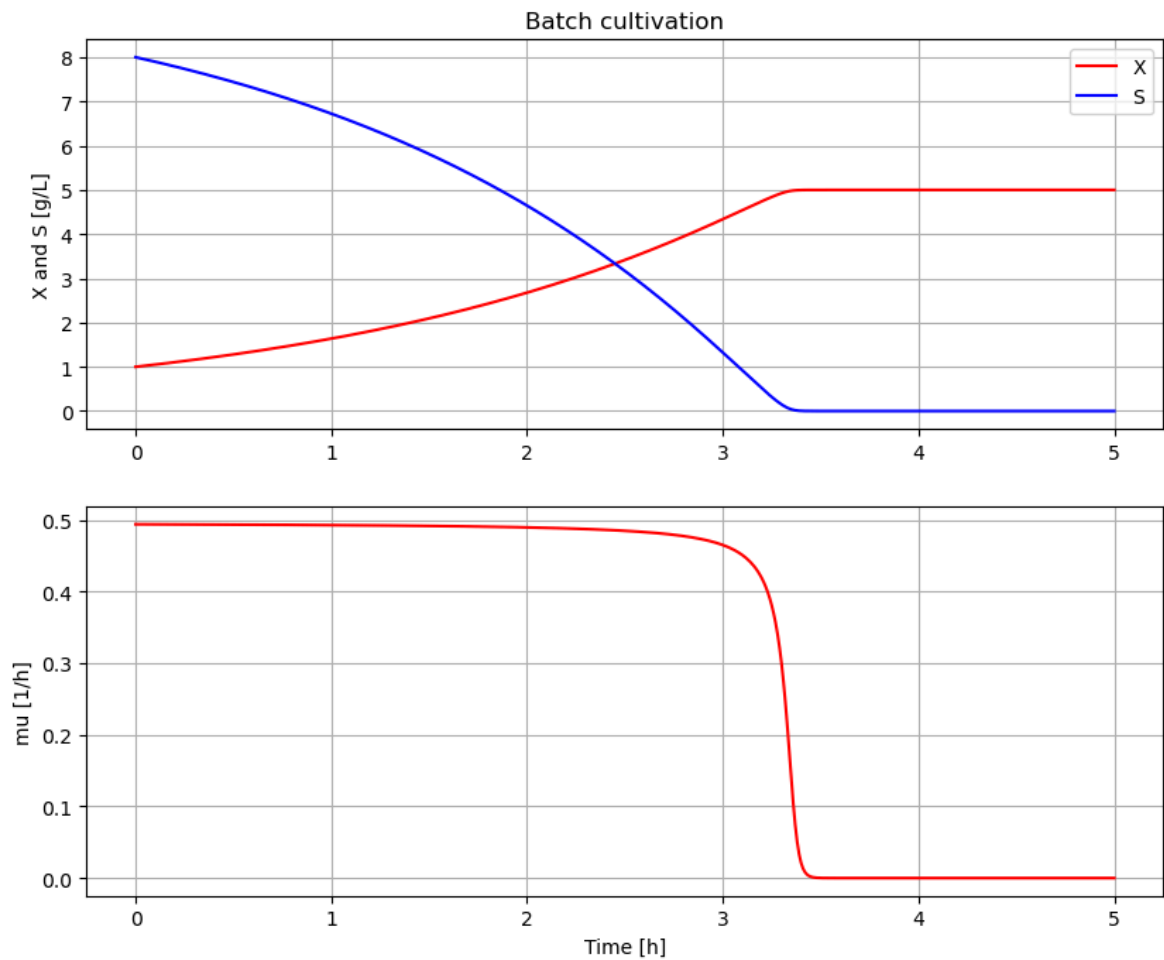
```
In [8]: describe('culture')
```

Simplified text book model - only substrate S and cell concentration X

```
In [9]: newplot(plotType='PhasePlane')
for value in [10, 12, 8]:
    init(VS_start=value)
    simu()
```

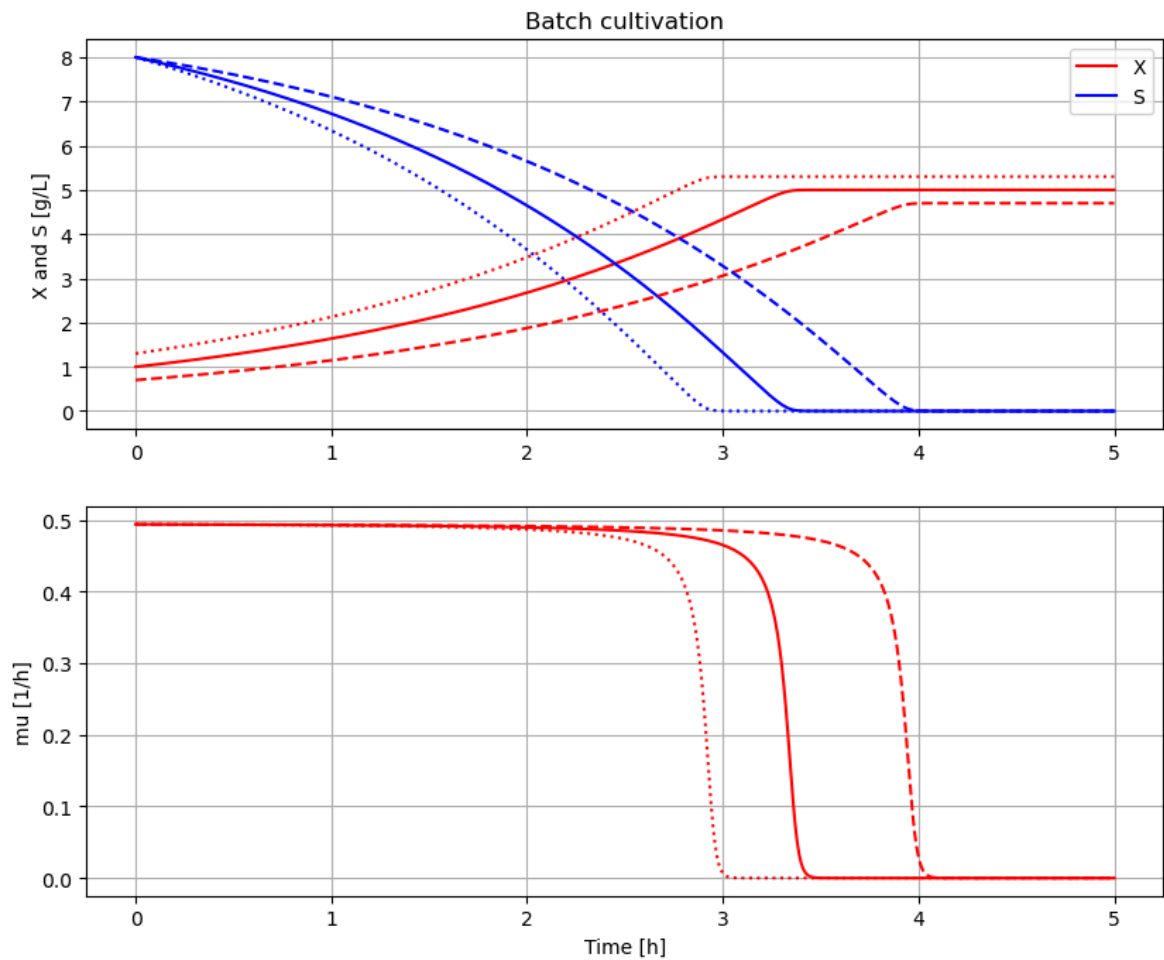


```
In [10]: # Simulation with default values of the process
newplot(plotType='TimeSeries')
simu()
```



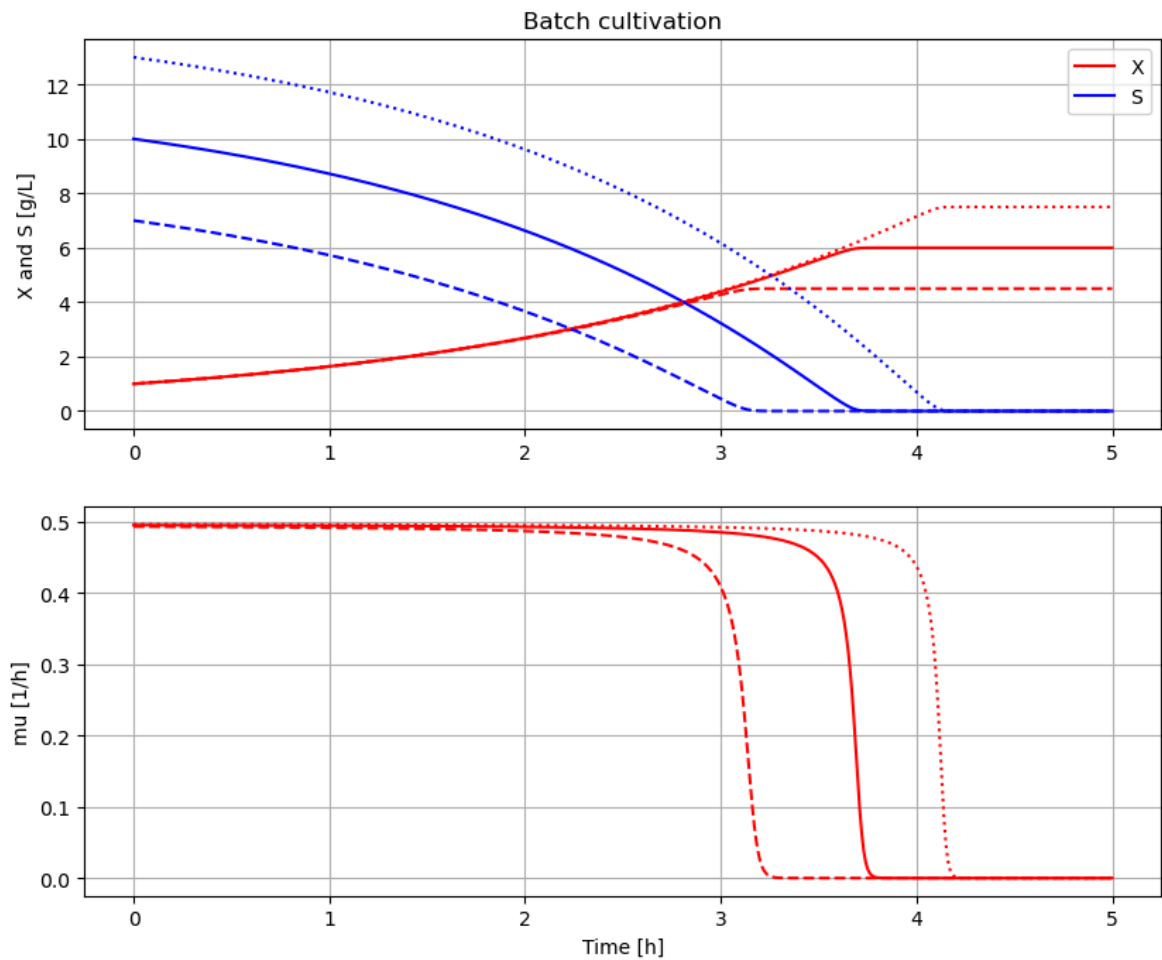
```
In [11]: # Simulation were initial value of biomass VX_start is varied
newplot(plotType='TimeSeries')
for value in [1.0, 0.7, 1.3]: init(VX_start=value); simu(5)

# Restore default value of VX_start
init(VX_start=1.0)
```



```
In [12]: # Simulation were initial value of substrate VS_0 is varied
newplot(plotType='TimeSeries')
for value in [10, 7, 13]: init(VS_start=value); simu(5)

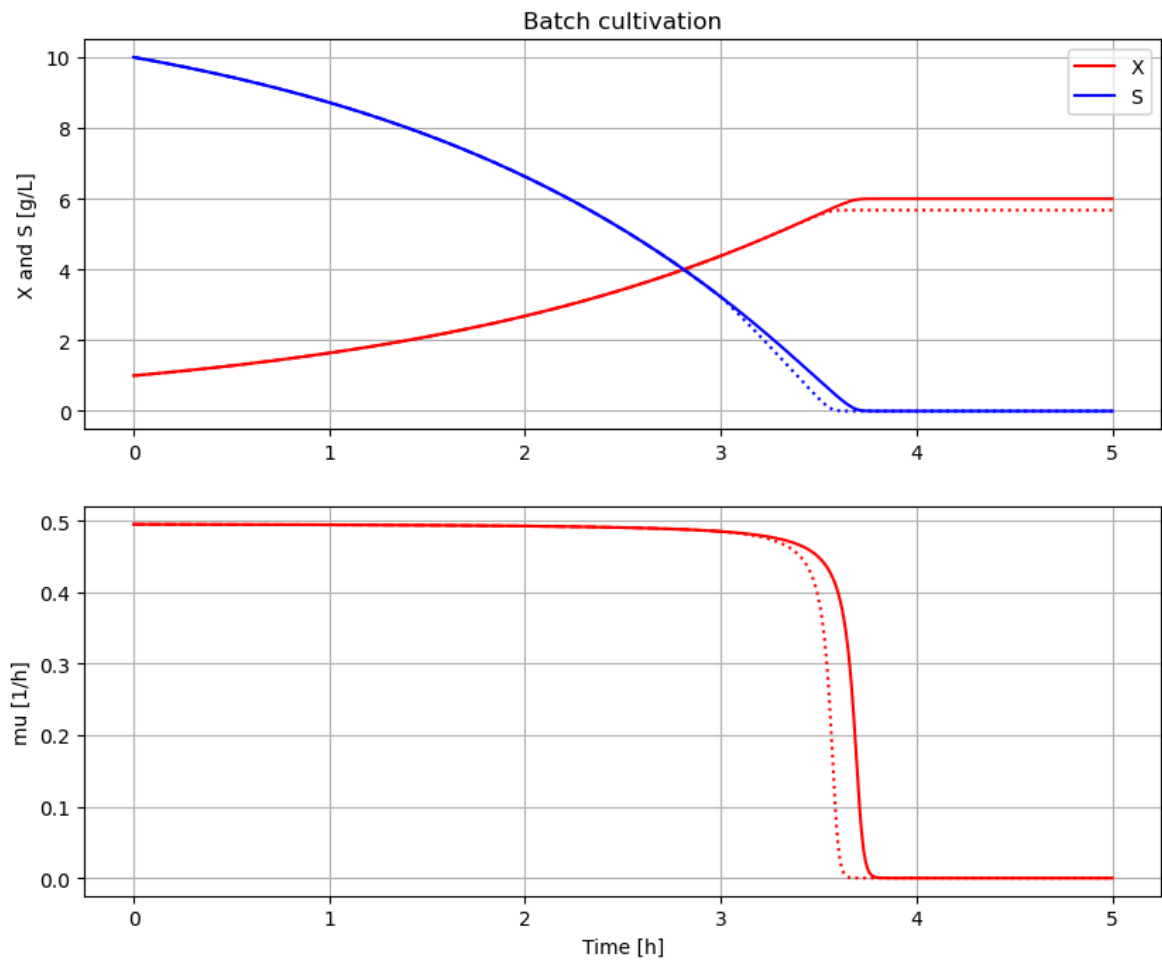
# Restore default value of VS_0
init(VS_start=10)
```



```
In [13]: # Simulation where metabolism is changed after 3 hours
newplot(plotType='TimeSeries')
simu(5)

simu(3)
par(Y=0.4, qSmax=1.0/(0.4/0.5)); simu(2, 'cont')

# Restore default value of Y and qSmax
par(Y=0.5, qSmax=1.0)
```



```
In [14]: describe('VX')
```

Substance mass : 5.677 [g]

```
In [15]: describe('V_start')
```

Initial reactor broth volume : 1.0 [L]

```
In [16]: describe('V')
```

Reactor broth volume : 1.0 [L]

```
In [17]: describe('MSL')
```

MSL: none

```
In [18]: system_info()
```

System information

- OS: Windows
- Python: 3.12.11
- Scipy: 1.16.0
- FMPy: 0.3.27
- FMU by: JModelica.org
- FMI: 2.0
- Type: CS
- Name: BPL.Examples_TEST2.Batch
- Generated: 2026-04-09T15:38:29
- MSL: 3.2.2 build 3
- Description: Bioprocess Library version 2.3.2
- Interaction: FMU_explore version 1.2.4

In [19]: `#!/lsb_release -a`

In [20]: `SDG('explain')`

"Soli Deo Gloria"

It is latin and means "To the honour of God".

The great composer Johan Sebastian Bach used to end his compositions with this small remark SDG.

And I like to do that too :).

In []: